

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

192311454

I Sanjana Shee.P (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Unit 1 : Assessment Tool 1

Analytical Problem Solving

Course Code : CSA1710 Artificial
Intelligence

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1). Delivery Drone as a Problem - Solving Agent :

Introduction :

An e-commerce company wants to use an autonomous drone to deliver packages across a city. The drone must determine the best route from the warehouse to the customer's location while avoiding obstacles, minimizing travel time and conserving battery power.

A goal-based problem-solving Agent is the most suitable agent for this task because it selects actions based on achieving a specific goal.

Problem Formulation :

Initial State : ~~Drone~~ is located at the warehouse.

Goal State : Package successfully delivered to the customer.

Actions : (*) Move North (*) Move East (*) Ascend /

Path cost : (*) Distance traveled
(*) Time taken
(*) Battery consumption.

Pseudocode :

function delivery_agent (Start, goal)

Open $\leftarrow$ Queue

Closed $\leftarrow$ Empty Set

Add Start to open

while Open is not empty Do

 current $\leftarrow$ remove node

 if current = goal then

 return Solution

 Add current to Closed

 for each neighbor of current Do

 if neighbor not in closed then

 add neighbor to open

return failure.

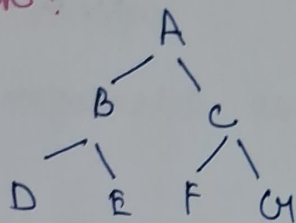
1. Iterative Deepening Search (IDS):

Introduction:

Iterative deepening search (IDS) is an uninformed search algorithm that repeatedly performs Depth-Limited Search (DLS) with increasing depth limits until the goal node is found.

IDS combines the advantages of Breadth-First Search (BFS) and Depth-First Search (DFS).

Example Graph:



Goal Node = F

Depth limit = 0

Only the root node is explored.

Visited nodes: A $\Rightarrow$ Goal not found.

Depth limit = 1

Nodes up to depth 1 are explored.

Visited nodes: A $\rightarrow$ B $\rightarrow$ C $\Rightarrow$ Goal not found.

Depth limit = 2

nodes up to depth 2 are explored.

Visited nodes: $A \rightarrow B \rightarrow D \rightarrow E \rightarrow C \rightarrow F$.

Goal node F is found.

Trace Table:

Iteration	Depth limit	Nodes Explored
1.	0	A
2.	1	A, B, C
3.	2	A, B, D, E, C, F

ID3 Algorithm:

function IDS (start, goal)

for depth = 0 to ∞ do

result $\leftarrow$ DLS (start, goal, depth)

If result = Success then

return solution.

End for.

3). Missionaries and Cannibals Problem :

Problem Statement :

Three missionaries and three cannibals must cross a river using a boat that can carry at most two people. At no time should cannibals outnumber missionaries on either bank when missionaries are present.

Problem Formulation :

State Representation :

It is represented as : (M, C, B)

where :

M = no. of missionaries

C = no. of cannibals

B = Boat Position

Initial State : $(3, 3, L)$

All missionaries and cannibals are on the left bank.

Goal State : $(0, 0, R)$

All missionaries and cannibals are on the right bank.

Actions :

Possible boat movements : (*) 2 Missionaries

(*) 2 Cannibals

(*) 1 Missionary

(*) 1 Cannibal

(*) 1 Missionary and 1 Cannibal .

Breadth - first Search Exploration :

Initial State :

$$S_0 = (3, 3, L)$$

Queue : $[(3, 3, L)]$

First State : $(3, 3, L)$

Valid Successor States : $(3, 1, R)$

$(2, 2, R)$

$(3, 2, R)$

Queue becomes : $[(3, 1, R), (2, 2, R), (3, 2, R)]$

Second State Explored : $(3, 1, R)$

Boat returns one Cannibal : $(3, 2, L)$

Queue becomes : $[(2, 2, R), (3, 2, R), (3, 2, L)]$

Third State Explored : $(2, 2, R)$